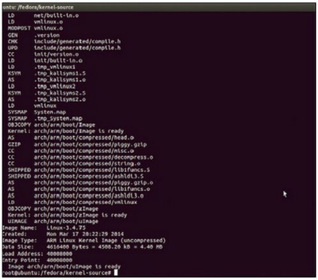

Figure 4: Snapshot of SBC OS kernel compilation - uImage

specific names in specific directory structures. Therefore, it is useful to follow the de facto standards that have emerged in Linux systems.

```
$ debootstrap --no-check-gpg --arch=armhf --foreign wheezy
```

The Init process and runlevels

In conventional Linux systems, Init is the first process started when a Linux kernel boots and it's the ancestor of all processes. Its primary role is to start appropriate service processes for the 'state' the system is to run in at boot and to shut down or start appropriate services if the system state changes (such as changing to the halt/shut down state). It can also create consoles and respond to certain types of events.

Init's behaviour is determined by its configuration file */etc/inittab*. Lines in */etc/inittab* have the following syntax:

```
id:runlevels:action:process
where:
id      –  1-4 (usually 2) character name for the line,
totally arbitrary;
runlevels  –  a list of runlevels the line applies to;
action     –  what init is to do and/or under what
conditions;
process    –  program/command to be run.
```

Here are the typical runlevels and what they mean for Red Hat family distros:

```
0 – halt system
1 – single user mode (no GUI)
2 – multiuser mode, no networking (no GUI)
3 – multiuser mode, networking (no GUI)
4 – unused
5 – multiuser mode (GUI/X11)   //FUTURE WORK
6 – reboot system
```

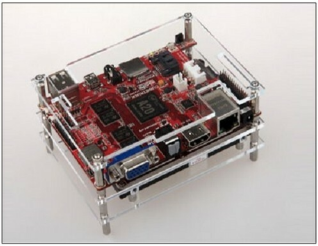
Figure 5: The Cubietruck SBC

A demo of the SBC OS and the single board computer called Cubietruck

FOSSOF 1.0 (Free and Open Source SOFtware) is a SBC OS that can be obtained from *https://github.com/gselvapravin/FossoF*. Developed by yours truly, it can be cloned by using the following commands:

```
$ sudo apt-get -y install git
$ cd ~
$ git clone https://github.com/gselvapravin/FossoF
$ chmod +x ./FossoF/fossof.sh
$ cd ./FossoF
$ ./fossof.sh
```

The compiled image will be located in */tmp/FossoF/output/debian_rootfs.raw.gz*. To write it to an SD card, decompress it and use Image Writer (Windows) or DD-it in Linux by using the following command:

```
$ dd bs=1M if FossoF_x.x_vga.raw of=/dev/sdx
```

Cubietruck is an SBC and is the third board of Cubieteam; so it is also called Cubieboard3. It's a new PCB model adopted with the Allwinner A20 main chip, just like Cubieboard2. But it is enhanced with some features, such as 2GB memory, an on-board VGA display interface, 1000M network interface, Wi-Fi+BT on board, support for Li batteries and RTC, and the SPDIF audio interface. END

By: Pravin Selva

The author is an Open Source enthusiast who wishes to share his knowledge in the true spirit of Open Source, and a Programmer Analyst Trainee at Cognizant Technology Solutions India Private Limited ("Cognizant").